Skills Summary

Solving Exponential Equations by Writing a Common Base

Use this reference while completing your worksheet or when studying.

Skill 1 — Equating Exponents When the Bases Match

The rule: for $b > 0$ with $b \neq 1$, the function b^u never repeats a value, so $b^u = b^v$ exactly when $u = v$. Matching bases is what *licenses* dropping them — with unequal bases the step is illegal.

Method: confirm the two bases are identical, set the exponents equal, solve that ordinary equation, then substitute back into both exponents to check.

Example: Solve $2^{x-3} = 2^{2x-7}$.
Step 1. Both sides are powers of the same base 2, so the equation is true exactly when the two exponents agree — no logarithm is needed.
Step 2. Set $x-3 = 2x-7$; subtracting x gives $-3 = x-7$, so $x = \mathbf{4}$. Check: both exponents equal 1, and $2^1 = 2^1$.

Try it: Solve $7^{4x} = 7^{x+6}$.

check: $\boldsymbol{x = 2}$

Skill 2 — Rewriting Both Sides Over a Common Base

When the bases differ, look for one base that generates both: $4 = 2^2$, $8 = 2^3$, $16 = 2^4$, $32 = 2^5$, $64 = 2^6$; $9 = 3^2$, $27 = 3^3$, $81 = 3^4$; $25 = 5^2$, $125 = 5^3$.

Two laws do all the work: $(b^m)^n = b^{mn}$ — a power of a power *multiplies* exponents — and $b^m b^n = b^{m+n}$ — a product *adds* them. A reciprocal is a negative exponent: $\frac{1}{2} = 2^{-1}$, $\frac{1}{16} = 2^{-4}$.

Rewrite, then equate exponents as in Skill 1. A fractional answer is normal.

Example: Solve $8^x = 32$.
Step 1. Neither side is a power of the other, but 8 and 32 are both powers of 2, so base 2 is the common ground that lets the exponents be compared.
Step 2. Rewrite: $(2^3)^x = 2^5$, that is $2^{3x} = 2^5$. Equating exponents gives $3x = 5$, so $x = \mathbf{\frac{5}{3}}$.

Try it: Solve $16^x = 8^{x+2}$.

check: $\boldsymbol{x = 6}$

Skill 3 — Exponential Models with a Common Base

The shape: a growth or decay model looks like $A = A_0\, b^{t/k}$ — starting amount A_0, growth factor b, one factor applied every k units of time.

Method: divide by the starting amount *first*, so the power stands alone; the coefficient is not part of the exponent and cannot be rewritten with it. Then write the isolated number as a power of the same base and equate exponents. Finally read the exponent as a count: $t/k = 4$ means four doublings or four half-lives have passed.

Its limit: this works only when the isolated ratio is a whole power of the base. It is not, logarithms are the general tool.

Example: A colony of 3 cells doubles hourly: $N = 3 \cdot 2^t$. When are there 96 cells?
Step 1. The 3 is a starting count, not part of the power, so it must be divided out before the two sides can be compared as powers of 2.
Step 2. From $3 \cdot 2^t = 96$, divide by 3 to get $2^t = 32 = 2^5$, so $t = \mathbf{5}$ hours — five doublings.

Try it: A 400 mg sample halves every 3 years: $M = 400\left(\frac{1}{2}\right)^{t/3}$. When is 25 mg left?

check: $t = \mathbf{12}$

(!) Watch out: $(2^3)^x$ is 2^{3x}, not 2^{3+x} — a power of a power multiplies. And never divide the exponents themselves: from $2^{4x} = 2^{3x+6}$ the next line is $4x = 3x + 6$, not $4x/3x = 6$.